M708 HW 03

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Exercise 2

It's proved in Voisin that (1) implies all the others. (2) implies (3) is clear: for $\alpha \in A^{1,0}$, if

$$\begin{aligned} d\alpha &= \partial\alpha + \bar{\partial}\alpha \\ &= d^{2,0}\alpha + d^{1,1}\alpha \\ &= d^{2,0}\alpha + d^{1,1}\alpha + d^{0,2}\alpha, \end{aligned}$$

then $d^{0,2}\alpha = 0$ as desired. Now we'll show that (3) and (4) each imply the Newlander-Nirenberg criterion $[T^{1,0}, T^{1,0}] \subseteq T^{1,0}$ for integrability of the almost complex structure J on X, from which we conclude that all the statements are equivalent.

Let $X, Y \in T^{1,0}$. To check that $[X, Y] \in T^{1,0}$, that is, $J[X, Y] = i[X, Y]$, it suffices to check that $\alpha(J[X, Y]) = \alpha(i[X, Y]) = i\alpha([X, Y])$ for all $\alpha \in A^1$. But since $\alpha^{1,0}(J[X, Y]) = i\alpha^{1,0}([X, Y])$ anyway, it suffices to check the above condition just for $\alpha \in A^{0,1}$, equivalently $\alpha^{0,1}(J[X, Y]) = -i\alpha^{0,1}([X, Y]) = i\alpha^{0,1}([X, Y])$, so it suffices to check $\alpha([X, Y]) = 0$ for all $\alpha \in A^{0,1}$.

Now for (3) $\implies$ (1), assume $d\alpha = \partial\alpha + \bar{\partial}\alpha$ for $(0, 1)$-forms α (by above and conjugation, this is equivalent to (3)). Then we use the formula

$$\begin{aligned} \alpha([X, Y]) &= X\alpha(Y) - Y\alpha(X) - d\alpha(X, Y) \\ &= -d\alpha(X, Y) \\ &= -\partial\alpha(X, Y) - \bar{\partial}(X, Y) \\ &= 0 \end{aligned}$$

since $\alpha(X) = \alpha(Y) = 0$ by the above reasoning since $X, Y \in T^{1,0}$ and $\alpha \in A^{0,1}$, and $\partial\alpha$ and $\bar{\partial}\alpha$ both have an antiholomorphic part, so the same reasoning applies to show that both send $X \wedge Y$ to 0. Thus, (3) $\implies$ (1).

Similarly, for (4) $\implies$ (1), assume $\bar{\partial}^2 = 0$, or equivalently $\partial^2 = 0$. Since above any point, all $(0, 1)$-forms are generated over C^∞ by those of the form $\bar{\partial}f$ for 0-forms f, it suffices to verify $\bar{\partial}f([X, Y]) = 0$. Now we compute:

$$\begin{aligned} \bar{\partial}f([X, Y]) &= df([X, Y]) - \partial f([X, Y]) \\ &= d^2f(X, Y) + df([X, Y]) - \partial f([X, Y]) \\ &= Xdf(Y) - Ydf(X) - \partial f([X, Y]) \\ &= X\partial f(Y) - Y\partial f(X) - \partial f([X, Y]) \\ &= \partial^2 f(X, Y) = 0 \end{aligned}$$

since $df = \partial f + \bar{\partial}f$ for 0-forms, and $\bar{\partial}f$ annihilates X and Y since they're holomorphic vector fields, and using the formula from above twice (one of them projected onto the $(0, 2)$ component). Therefore, (4) $\implies$ (1), and thus all the statements are equivalent.

Exercise 9

On closed forms $\omega \in A^{1,1}(X) \cap A^2_{\mathbb{R}}(X)$, the only remaining condition to be Kähler is positivity: $\omega(v, Jv) > 0$ for all $v \in T_{X,x}$, for all $x \in X$. The "convex cone" part is immediate: it is clear that any positive linear combination of such ωs will still satisfy the positivity condition. It remains to check openness of this cone in a reasonable topology.

All we'll require of the topology is that it refines the one given by the following norm. On each tangent space $T_{X,x}$, ω is given by a skew-symmetric real matrix A_x, so the eigenvalues of $A_x J$ are nonnegative real and vary continuously, so since X is compact, we can take the maximum over all of them as x ranges over X, and define this to be the norm of ω. It's easy to verify that this is a norm, using the triangle inequality for maximum eigenvalues λ, μ of square matrices A, B: (which we will use again below): the maximum eigenvalue of $A + B$ is at most $\lambda + \mu$.

Now for a Kähler form ω on X, we claim there is a ball in this norm which is comprised of Kähler forms. The radius ϵ of the ball is the minimum over X of the minimum eigenvector of $A_x J$, in the notation used above. Then, using the same triangle inequality as above but for the minimum eigenvalue of ω, we have, for any $\theta \in A^{1,1}(X) \cap A^2_{\mathbb{R}}(X)$ with $d\theta = 0$ and norm at most ϵ, $\omega + \theta$ is still Kähler, since on each tangent space $T_{X,x}$, we have that the minimum eigenvalue of $\omega + \theta$ is at least $\lambda - \mu > 0$ since $\mu < \epsilon = \min_X \lambda$, where λ is the minimum eigenvalue of $A_x J$ and μ is the maximum eigenvalue of $B_x J$, with A_x, B_x the matrices representing ω and θ, respectively. Thus $\omega + \theta$ satisfies the positivity criterion on each tangent space, and hence does so for the entire compact manifold X, hence is Kähler. Therefore, the set of Kähler forms forms an open convex cone in the space $\{\omega \in A^{1,1}(X) \cap A^2_{\mathbb{R}}(X) : d\omega = 0\}$.

Exercise 11

We have the explicit form in one chart (covering almost all of $\mathbb{CP}^n$)

$$\omega = \frac{i}{2\pi} \frac{\left(1 + \sum_i |z_i|^2\right) \sum_i dz_i \wedge d\bar{z}_i - \left(\sum_i \bar{z}_i dz_i\right) \wedge \left(\sum_i z_i d\bar{z}_i\right)}{\left(1 + \sum_i |z_i|^2\right)^2}$$

We change to polar coordinates in each component: $z_i = r_i e^{i\theta_i}$, so $|z_i|^2 = z_i \bar{z}_i = r_i^2$, and

$$dz_i = e^{i\theta_i} dr_i + i r_i e^{i\theta_i} d\theta_i = r_i^{-1} z_i dr_i + i z_i d\theta_i,$$

and accordingly $d\bar{z}_i = r_i^{-1} \bar{z}_i dr_i - i \bar{z}_i d\theta_i$. Thus

$$\begin{aligned} dz_i \wedge d\bar{z}_i &= (r_i^{-1} z_i dr_i + i z_i d\theta_i)(r_i^{-1} \bar{z}_i dr_i - i \bar{z}_i d\theta_i) \\ &= -i r_i^{-1} z_i \bar{z}_i dr_i \wedge d\theta_i - i z_i r_i^{-1} \bar{z}_i dr_i \wedge d\theta_i \\ &= -2 i r_i dr_i \wedge d\theta_i \end{aligned}$$

and

$$\bar{z}_i dz_i = r_i^{-1} z_i \bar{z}_i dr_i + i z_i \bar{z}_i d\theta_i = r_i dr_i + i r_i^2 d\theta_i$$

Plugging all this in, we get

$$\omega = \frac{i}{2\pi} \frac{-2i\left(1 + \sum_i r_i^2\right) \sum_i r_i dr_i \wedge d\theta_i - \left(\sum_i r_i dr_i + i r_i^2 d\theta_i\right) \wedge \left(\sum_i r_i dr_i - i r_i^2 d\theta_i\right)}{\left(1 + \sum_i r_i^2\right)^2}$$

In the latter product of 1-forms, only the cross-terms survive since the square of any 1-form is 0, so we are left with $2i \sum_{i,j} r_i r_j^2 dr_i \wedge d\theta_j$. The diagonal terms here are of the form $r_i^3 dr_i \wedge d\theta_i$. In the former product of the scalar with 2-forms, the diagonal terms are also of the form $r_i^3 dr_i \wedge d\theta_i$, but of the opposite sign, so they cancel. Also cancelling the $-2i$, we are left with

$$\omega = \frac{1}{\pi} \frac{\left(\sum_i (1 + \sum_{j \neq i} r_j^2) r_i dr_i \wedge d\theta_i\right) - \sum_{i \neq j} r_i r_j^2 dr_i \wedge d\theta_j}{(1 + \sum_i r_i^2)^2}$$

Now we take this to the nth exterior power. Note that the expression above is explicitly a linear combination of terms $dr_i \wedge d\theta_j$ for $1 \le i, j \le n$. We might as well arrange the coefficients into a matrix:

$$M = \begin{bmatrix} (S - r_1^2)r_1 & -r_1 r_2^2 & -r_1 r_3^2 & \cdots & -r_1 r_n^2 \\ -r_2 r_1^2 & (S - r_2^2)r_2 & -r_2 r_3^2 & \cdots & -r_2 r_n^2 \\ -r_3 r_1^2 & -r_3 r_2^2 & (S - r_3^2)r_3 & \cdots & -r_3 r_n^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -r_n r_1^2 & -r_n r_2^2 & -r_n r_3^2 & \cdots & (S - r_n^2)r_n \end{bmatrix}$$

where $S = 1 + \sum_{i=1}^n r_i^2$. Furthermore, note that the only terms that survive in the wedge product are those coming from a permutation $\sigma \in S_n$, in the sense that we wedge together $dr_i \wedge d\theta_{\sigma(i)}$ for $i = 1, \dots, n$, since we must choose n 2-forms to wedge together, and if we pick any two in the same row or column above, they will have two of some 1-form, and hence vanish. Since 2-forms commute with one another, we just get $n!$ copies of the same term from each permutation σ, so we'll bring that coefficient out front in a moment.

From a permutation σ, the differential n-form we obtain (with some coefficient) is

$$\begin{aligned} dr_1 \wedge d\theta_{\sigma(1)} \wedge dr_2 \wedge d\theta_{\sigma(2)} \wedge \cdots \wedge dr_n \wedge d\theta_{\sigma(n)} &= \operatorname{sgn}(\sigma) dr_1 \wedge d\theta_1 \wedge dr_2 \wedge d\theta_2 \wedge \cdots \wedge dr_n \wedge d\theta_n \\ &= \operatorname{sgn}(\sigma) dV \end{aligned}$$

since swapping two $d\theta_i$ terms involves moving one of them over two 1-forms, then the other over one 1-form, for a net sign change of -1, meaning transpositions correspond to signs. Therefore, using Leibniz's formula for the determinant, we see that in fact

$$\omega^n = \frac{n!}{\pi^n} \frac{\det(M)}{S^{2n}} dV$$

Now we must compute $\det(M)$. First, let's factor out $\prod_i r_i$, since the ith row is a multiple of r_i. Then, we can factor out $\prod_i r_i^2$, taking out r_i^2 from the ith column. We're left with:

$$N = \begin{bmatrix} S - r_1^2 & -r_2^2 & \cdots & -r_n^2 \\ -r_1^2 & S - r_2^2 & \cdots & -r_n^2 \\ \vdots & \vdots & \ddots & \vdots \\ -r_1^2 & -r_2^2 & \cdots & S - r_n^2 \end{bmatrix}$$

We note now that this is the sum of a simpler matrix N', whose ith column consists of all $-r_i^2$s, with the scalar matrix S. Since these are simultaneously diagonalizable, we obtain the eigenvalues of N by adding S to each of the eigenvalues of N'. We easily see that N' has one nonzero eigenvalue, which is $\sum_i -r_i^2 = 1 - S$, and the rest are 0. Thus, the eigenvalues of N are 1, and S with multiplicity $n - 1$. Therefore, $\det(M) = r_1 r_2 \cdots r_n \cdot S^{n-1}$, so we have

$$\omega^n = \frac{n!}{\pi^n} \frac{r_1 r_2 \cdots r_n \cdot S^{n-1}}{S^{2n}} dV = \frac{n!}{\pi^n} \frac{r_1 r_2 \cdots r_n}{S^{n+1}} dV$$

Now, let's just compute the integral. First, we can integrate out all the $d\theta_i$s, acquiring a factor of $(2\pi)^n$ and leaving us with

$$\int_{\mathbb{CP}^n} \omega^n = n! 2^n \int_{\mathbb{R}^n_{>0}} \frac{r_1 \cdots r_n}{(1 + r_1^2 + \cdots + r_n^2)^{n+1}} dr_1 \cdots dr_n$$

Let's just do the dr_n integral, and substitute $u = 1 + r_1^2 + \cdots + r_n^2$, so that $du = 2r_n dr_n$. Then we have

$$\begin{aligned}
\int_{\mathbb{CP}^n} \omega^n &= n!2^{n-1} \int_{\mathbb{R}^{n-1}_{>0}} r_1 \cdots r_{n-1} \int_{1+r_1^2+\cdots+r_{n-1}^2}^{\infty} \frac{du}{u^{n+1}} dr_1 \cdots dr_{n-1} \\
&= n!2^{n-1} \int_{\mathbb{R}^{n-1}_{>0}} r_1 \cdots r_{n-1} \left[-\frac{1}{n} \frac{1}{u^n} \right]_{1+r_1^2+\cdots+r_{n-1}^2}^{\infty} dr_1 \cdots dr_{n-1} \\
&= (n-1)!2^{n-1} \int_{\mathbb{R}^{n-1}_{>0}} r_1 \cdots r_{n-1} \frac{1}{(1 + r_1^2 + \cdots + r_{n-1}^2)^n} dr_1 \cdots dr_{n-1}
\end{aligned}$$

By induction, we clearly see that this reduces to the integral for the $n = 1$ case, which we already know is 1. Therefore, we conclude that

$$\int_{\mathbb{CP}^n} \omega^n = 1$$

as desired.

Exercise 15

Let r be the rank of E. A connection ∇ on E is an $\mathbb{R}$-linear map $A^0(E) \to A^1(E)$ satisfying the Leibniz rule $\nabla(f\sigma) = \sigma \otimes df + f\nabla(\sigma)$ for $f \in C^\infty(X)$, $\sigma \in A^0(E)$. The induced connection is then an $\mathbb{R}$-linear map $A^0(\bigwedge^r E) \to A^1(\bigwedge^r(E))$ satisfying the Leibniz rule. On a generic element $\sigma_1 \wedge \cdots \wedge \sigma_r$ with $\sigma_i \in A^0(E)$, this is:

$$\nabla(\sigma_1 \wedge \cdots \wedge \sigma_r) = \sum_{i=1}^{r} \sigma_1 \wedge \cdots \wedge \nabla(\sigma_i) \wedge \cdots \wedge \sigma_r$$

For the rest of the problem, fix a trivialization $E|_U \cong U \times \mathbb{C}^r$ with basis $e_1, \ldots, e_r$. In our trivialization, the line bundle $\det(E)$ is generated over $C^\infty(U)$ by $e_1 \wedge \cdots \wedge e_r$.

If ∇ on $E|_U$ has connection matrix $A \in A^1(X)^* \otimes A^1(X)$, so $\nabla(f) = df + Af$ where $f = (f_1, \ldots, f_r)$ are C^∞ functions representing a section of E in our trivialization, then on $\det(E)$, we have

$$\begin{aligned}
\nabla(g \cdot e_1 \wedge \cdots \wedge e_r) &= e_1 \wedge \cdots \wedge e_r \otimes dg + g\nabla(e_1 \wedge \cdots \wedge e_r) \\
&= e_1 \wedge \cdots \wedge e_r \otimes dg + g \sum_{i=1}^{r} e_1 \wedge \cdots \wedge \nabla(e_i) \wedge \cdots \wedge e_r \\
&= e_1 \wedge \cdots \wedge e_r \otimes dg + g \sum_{i=1}^{r} e_1 \wedge \cdots \wedge \left(\sum_{j=1}^{r} A_{ji} e_j \right) \wedge \cdots \wedge e_r \\
&= e_1 \wedge \cdots \wedge e_r \otimes dg + g \sum_{i=1}^{r} e_1 \wedge \cdots \wedge A_{ii} e_i \wedge \cdots \wedge e_r \\
&= e_1 \wedge \cdots \wedge e_r \otimes dg + \left(\sum_{i=1}^{r} A_{ii} \right) g \cdot e_1 \wedge \cdots \wedge e_r
\end{aligned}$$

so we see that the connection matrix of the induced connection on $\det(E)$ is $\mathrm{tr}(A)$. Further, by the general formula $F_A = dA + A \wedge A$ for curvature of a connection, we see that the curvature of the induced connection is $F_{\mathrm{tr}(A)} = d\,\mathrm{tr}(A) + \mathrm{tr}(A) \wedge \mathrm{tr}(A) = \mathrm{tr}(dA)$, since the square of any 1-form (such as $\mathrm{tr}(A)$) is 0 and d is linear. Further, this coincides with the trace of the curvature F_A of ∇ on E, since $\mathrm{tr}(F_A) = \mathrm{tr}(dA + A \wedge A) = \mathrm{tr}(dA)$, since $\mathrm{tr}(A \wedge A) = \sum_{ij} A_{ij} A_{ji} = 0$ since swapping the indices permutes the summands but swaps the factors, and 1-forms anticommute.